Physics Labs

Alveary

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Science

Physics Labs

GRADE(S): 11-12

About the Course

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method. Labs for this course are integrated into the lessons to facilitate adequate time for more involved activities and to better coordinate with the lessons.

This topic is included in the following course(s): Science: Physics

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
11-12	Physics Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Physics				
Physics Lessons Nature Walks & Scouting: Grades 9-12	Physics Lessons	Physics Lessons	Physics Lessons Nature Notebook: Grades 9-12	Physics Lessons Physics Labs

Planning & Prep

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LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Physics Labs

☐ Preview science labs and purchase materials as indicated on the supply list. Print or shortcut any links, as desired.

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

Physics Labs

How To Approach

Introduce

• This course uses more open-ended 'projects' for lab, but every activity still has the same flow, which follows the scientific method.

• Learners begin with an introduction. How does the project relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is how hypotheses are formed.

• Learners are not given a Procedure and must decide for themselves how to approach the lab. If they are unsure what to do, it can help to dialogue with someone as they work through the ideas.

• Once they have had a chance to think, learners compose a prelab narration to put these introductory ideas, hypotheses, and plans into their lab notebooks. These prelab narrations may seem short and even incomplete if learners are new to keeping a lab notebook, but that is okay.

• If learners have difficulty or are easily frustrated, then consider allowing a digital notebook for typing, a scribe, or the use of assistive technology, as appropriate.

• After they complete their prelab narration, learners should collect any materials, being sure to let teachers know if something is missing or to remind the teacher if something needs to be purchased at the store.

Lab Procedure

• Learners complete the activity.

• They should use their lab notebooks to record what they do and capture any data, which can be printed from the computer and taped into the notebook.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. Similar to the prelab narration, the concluding or postlab narration is a chance to think about and put into words. This time, learners are considering what they learned from the lab and what they could learn more about if they were to continue.

• Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

• Depending on the interest of the learner and the priorities of the teacher, the learner might be encouraged to spend more time on those ideas of what more they could learn or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.